

SPENCER DELCORE

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OBJECTIVE

University of Waterloo Computer Engineering graduate looking to pursue a career in Machine Learning to foster knowledge growth and to better contribute to the field of Artificial Intelligence.

WORK EXPERIENCE

Molex. - *Embedded Software Developer*
Waterloo, ON

May. 2019 - Present

- Involved in designing a platform agnostic embedded run-time framework written in C
- Key implementer of run-time framework for industrial network protocol applications
- Technical evangelist to promote and support the framework through out-growth of work
- Worked with R&D Manager to evaluate Machine Learning solutions and TensorFlow in Python for various industrial applications

Molex. - *Co-op Developer*
Waterloo, ON

Sept. 2018 - Dec. 2018

- Responsible for evaluating QNX Hypervisor on an Intel NUC and Renesas R-Car-H3
- Built various Linux images using both Yocto and Buildroot, along with sample programs in C
- Developed an IoT Edge Gateway for PROFINET enabled devices using Azure IoT Edge
- Designed a Human Machine Interface (HMI) connected to an OPC Server

Evertz Microsystems Ltd. - *Embedded Systems Design Engineering, Co-op* Sept. 2017 - Dec. 2017
Burlington, ON

- Developed a multi-threaded framework for generating embedded user interfaces using Qt and QML
- Implemented an embedded SQLite database, and configured it for performance
- Communicated through a defined protocol with various systems via D-Bus
- Parsed multiple JSON structures to generate various user interfaces

Cognitive Systems Corp. - *Augmented Reality (AR) Developer, Co-op*
Quantum Valley Investments, Waterloo, ON

Jan. 2017 - April. 2017

- Responsible for creating an AR demonstration to visualize wireless signals using a Microsoft HoloLens
- Dynamically generated interactive holograms using Unity and C#
- Created a Python program that interfaces with Cognitive's system on a chip (SoC) to detect wireless signals
- Implemented websockets to communicate between the SoC and the HoloLens

EDUCATION

University of Waterloo, Waterloo, Ontario

2014 - 2019

Bachelor of Applied Science (with distinction), Honours Computer Engineering - Co-operative Program

TECHNICAL SKILLS

Professional Experience

C/C++

Familiar With

Python, JavaScript, SQL

Frameworks/Technologies

TensorFlow, Azure IoT Edge, Qt, AWS

Continuing Education

Coursera, deeplearning.ai 'Specialization'

PROJECTS

Video Event Notifier - *C, Python, JavaScript, MySQL*

2019

- A micro-service approach to receive notifications for events during a video
- MQTT messages are sent between a MySQL database application (C), a video monitoring service (Python), and a notification service (JavaScript)
- The video service monitors a video and provides the database application the elapsed time, the SQL application then sends an event to the notification service
- Each service is unaware of the architecture of the other so that both the video service and the notification service are replaceable

Home Server - *Docker, Jenkins, OpenVPN, JupyterLab, pfSense*

2019

- Created a home server to host various applications for personal use managed through Docker
- Hosted a JupyterLab instance for building Tensorflow models with a GPU
- Set up a Jenkins instance to build a number of projects hosted through git
- Uses pfSense as a router to ensure security and have a single point of entry through an OpenVPN server instance

Terrainer - *Arduino, React Native*

2019

- Retrofitted a treadmill with two Arduino cores (ATMega328) to receive speed and incline commands from an Android/iOS application over Bluetooth
- Allows users to input real world routes through Google Maps API to simulate on the treadmill
- Provided real-time statistics based on the current user and treadmill status

VHDL Compiler - *Java, VHDL*

2017

- Created a VHDL synthesizer and simulator using Java
- Using VHDL (combinational circuits only) as the input language, circuit diagrams and simulations are produced
- Gathered important compiler concepts such as regular languages, regex, DFA, lexing, recursive decent parsing, and context-free grammars

Room Sensors - *C++, Python, Bash, JavaScript*

2017

- Plots information stored in a database containing data from Arduino sensors
- Communicate and store sensor data from an Arduino to a Raspberry Pi using a created serial protocol
- Chart data from JSON reports received from a Raspberry Pi through websockets
- Created an install script that sets up the websocket, web page, cron job, Arduino, and MySQL database